Huaixiang LI

Phone: +86 15287649692 | **Email:** huaixianglee@gmail.com
Google Scholar: Huaixiang Li - Google Scholar | **Address:** Kunming, China

EDUCATION

Kunming University of Science and Technology (KUST) **2023 – 2027**
Bachelor's Degree in Chemical Engineering and Technology **GPA: 3.62/4.0; Average Score: 87.4/100; Rank: 3/109**
Relevant Courses: Principles of Chemical Engineering, Chemical Engineering Thermodynamics, Chemical Process Modeling, Basis of Industrial Catalysis, Instrumental Analysis, Course Design for Chemical Engineering

RESEARCH INTERESTS

Keywords: non-precious metal catalysts, defect engineering, propane dehydrogenation (PDH), CO_2 hydrogenation

- Designing non-precious-metal catalysts for selective light-alkane dehydrogenation through defect engineering, alloying, and metal-support interaction regulation.
- Extend these strategies to CO_2 valorization, specifically targeting CO_2-assisted propane dehydrogenation (CO_2-ODHP) and CO_2 hydrogenation to methanol, to develop catalysts that efficiently activate inert C-H and CO_2 bonds.

PUBLICATIONS

Li, H., Zhang, T., Tao, Z., Lin, Y., Zhou, S., Wang, H., Li, Y., He, D., & Luo, Y. **Stabilizing Ni^{2+} sites via boron-induced structural modulation for propane dehydrogenation.** *Catalysis Science & Technology,* **2026**. DOI: 10.1039/D6CY00466K.

- Led catalyst design, synthesis, catalytic evaluation, characterization, and manuscript preparation.
- Boron-induced interfacial reconstruction stabilizes highly dispersed Ni^{2+} species and suppresses NiO crystallite growth and aggregation.
- Modified electronic structure weakens coke condensation, enhancing reactivity and stability in propane dehydrogenation.

Tao, Z., **Li, H.**, He, D., Huang, Z., Liang, J., Dong, J., Cao, X., & Luo, Y. **$Ga^{\delta+}$-H active sites-driven reverse water gas shift reaction with high CO selectivity.** *Chemical Engineering Journal,* **2026**. DOI: 10.1016/j.cej.2026.175788.

- Contributed to catalytic performance evaluation and in situ DRIFTS analysis, supporting mechanistic investigation of $Ga^{\delta+}$-H active species during the RWGS reaction.
- First identification of $Ga^{\delta+}$-H species as the key active sites for reverse water-gas shift (RWGS) on Ga/Al_2O_3, enabling rapid CO_2-to-formate transformation.
- Achieves near-100% CO selectivity from 300 °C to 700 °C with exceptional RWGS activity.

Zhang, T., He, D., Huang, Z., **Li, H.**, Liang, J., Dong, J., Zhou, W., Li, Y., & Luo, Y. **Controlled lattice-oxygen release at Ce–O–Ni interfaces enable coke and sulfur resistance in dry reforming of methane.** *Submitted to Nature Catalysis,* **2026**.

- Contributed to catalytic evaluation, physicochemical characterization, data analysis, and mechanistic interpretation of coke and sulfur resistance under DRM conditions.
- Ce-O-Ni interfacial sites maintain electron-deficient Ni species and regulate lattice-oxygen release from CeO_2, moderating CH_4 dissociation and suppressing coke formation.
- CO_2-derived formate intermediates promote the removal of H_2S-derived sulfur through a sulfite-to-SO_2 pathway, enabling strong H_2S resistance and a low deactivation constant (kd = 0.0016 h^{-1}).

RESEARCH EXPERIENCE

Research on Multi-functional Catalysts for the Stepwise Catalytic Hydrogenation of CO_2 to Aviation Kerosene
Researcher | Supervisor: Dr. Yubing Li **Jan. 2026 – Present**
A Funded Project Supported by the National Natural Science Foundation of China Young Scientists Fund

- Developing MXene-based catalysts for CO_2 hydrogenation to methanol, focusing on interfacial-site regulation and the activation and selective conversion of CO_2.
- Designing LDH-derived catalysts for CO_2-assisted propane dehydrogenation and investigating composition–structure–performance relationships through catalytic testing and physicochemical characterization.

High-Efficiency Application and Performance Optimization of Defect-Controlled Metal Oxide Catalysts in Propane Dehydrogenation
Researcher and Project Lead | Supervisor: Professor Dedong He **Apr. 2025 – Present**
A Funded Project Supported by the Yunnan Provincial University Students' Innovation and Entrepreneurship Fund

- Designed high-performance Ni-based catalysts for PDH through strategic defect engineering and acidity modulation of catalyst supports.
- Led the entire research pipeline, from experimental design and catalyst synthesis to systematic performance evaluation and in-depth data analysis, ensuring the project aligned with its objectives.
- Explored the relationship between the structure and the activity of the catalyst. Mastered and applied advanced characterization techniques, including XRD, XPS, H_2-TPR, Raman, UV-Vis, EPR, and in situ DRIFTS, to analyze the catalyst structure, metal valence state, oxygen vacancy defects, and reaction mechanism.
- Achieved exceptional catalytic performance with the developed Ni-based catalyst, maintaining >93% selectivity to propylene and >30% propane conversion at 600°C, surpassing the performance of most Ni-based catalysts reported in the literature.

Silicon-Iron Purification: Waste-to-Green Energy Transformation using Advanced Functional Materials for Water Purification
Core Member | Supervisor: Professor Dedong He **Sep. 2024 – Apr. 2026**
China International University Students' Innovation Competition

- Transformed industrial dangerous waste into high-performance ferrosilicon-based water purification materials, achieving a green circular economy model of treating waste with waste.
- Mastered catalyst material preparation and led systematic testing and characterization to correlate material structure with performance for optimization.

Research on Environment-Responsive Polysaccharide Nanohydrogels with Hydrophobic Microdomains as a Carrier System
Researcher | Supervisor: Professor Hongying Su **Sep. 2023 – Apr. 2024**
A Funded Project Supported by the National Natural Science Foundation of China
- Conducted literature review and laboratory training on polysaccharide nanohydrogels for stimulus-responsive drug delivery.

Preliminary Study on Structural Compatibility between Plant Purple Pigments and Silk/Cotton Fibers
Core Member | Supervisor: Associate Professor Donghui Liu **Apr. 2024 – Apr. 2025**
A Funded Project Supported by the Yunnan Provincial University Students' Innovation and Entrepreneurship Fund
- Designed dyeing experiments and analyzed pigment-fiber interactions in silk and cotton materials.

PROFESSIONAL EXPERIENCES

AI for Science Institute, Beijing **Mar. 2025 – Present**
Data Annotator
- Contributed to the training of the Uni-Miner project, which is a large language model. Performed high-quality annotation and proofreading of massive amounts of scientific literature and patent data to build robust training datasets.
- Reviewed the results of the automatic annotations of the AI model, accurately identified and corrected errors, such as entity recognition and relationship extraction, effectively improving the quality and reliability of the dataset.

Online Tutoring Initiative **Jan. 2025 – Present**
Founder & Tutor
- Leveraged prior tutoring experience to independently launch and manage a private online tutoring service, assuming full responsibility for curriculum development, student acquisition, and session delivery.
- Orchestrated all operational aspects of the initiative, from market outreach and client consultations to scheduling and session delivery, demonstrating strong entrepreneurial initiative.
- Developed a stable client base by developing and implementing long-term, personalized learning schemes for over 15 students, transitioning the service from ad-hoc support to a comprehensive knowledge system building.

ADDITIONAL ACADEMIC TRAINING

Participant | 2026KUST-Cambridge Academic Research Program **May 2026 – Aug. 2026**
- Completed an AI-focused academic program involving lectures by University of Cambridge faculty and discussions on machine learning applications in scientific research and engineering.
- Participated in academic discussions on AI-assisted experimental design and chemical process optimization.

Participant | Peking University Guanghua Initiative **Sep. 2022 – Feb. 2023**
- Attended a series of structured lectures covering cutting-edge topics and fundamental theories in economics and finance. Applied learnings to explore the crucial role of these two fields in driving social development and resource allocation.
- Reflected on the connection between individual agency and social responsibility; further cultivated abilities to conduct logical analysis and strategic thinking from a macro perspective.

EXTRACURRICULAR ACTIVITIES

Volunteer | United Nations Children's Fund **Feb. 2021 – Present**
- Participated in research and advocacy work on the living conditions of children worldwide, gaining a deeper understanding of the core challenges in education and health faced by children across different countries.
- Collaborated within a multicultural team of young volunteers from diverse backgrounds. Gained insights into global social issues and improved cross-cultural communication skills.

AWARDS AND HONOURS

Competition Awards
- **National Second Prize,** The 20th National College Students Chemical Engineering Design Competition. **2026**
- **Outstanding Poster Award,** 13th National Symposium on Catalyst Preparation Science and Technology **2026**
- **Third Prize,** the 8th National University Students' Chemical Engineering Experiment Competition (Southwest China Division) **2025**
- **Third Prize,** the Contemporary Undergraduate Mathematical Contest in Modeling (Yunnan Provincial Division) **2024**
- **Third Prize,** the 4th Yunnan Provincial University Students' Energy Conservation & Emission Reduction Competition **2025**
- **Silver Award,** the China International University Students' Innovation Competition (University Level) **2025**

Scholarships & Honours
- **Provincial Government Scholarship** **2026**
- **Provincial "Five Merits" Student** (For Achieving Excellence in Moral, Intellectual, Physical, Aesthetic, and Labor Education) **2026**
- **He Dong Scholarship**, KUST **2024**
- **First-class Scholarship**, KUST **2025 & 2026**
- **Second-class Scholarship**, KUST **2023 & 2024**
- **Innovation and Entrepreneurship Award**, KUST **2025 & 2026**

ADDITIONAL INFORMATION

Technical Skills
- **Catalyst Synthesis:** Impregnation, co-precipitation, ion exchange, catalyst calcination and activation
- **Catalytic Evaluation:** Fixed-bed reactor operation, gas-phase catalytic testing, product analysis by GC
- **Characterization & Analysis:** XRD, XPS, H_2-TPR, NH_3-TPD, O_2-TPO, in situ DRIFTS, UV-Vis, Raman
- **Software:** Origin, ChemDraw, Diamond, Cinema 4D, HSC

Languages: Chinese (native), English (TOEFL 5.0)